

State as Computation: Moving Structural Search between Representation Construction and Downstream Reasoning

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Abstract

Reasoning systems are commonly compared as if the state presented to the downstream learner were a fixed observation. We study a different resource boundary: **constructing task-relevant state is itself computation**, and this computation can substitute for structural search performed downstream. For a query family F , any fixed representation supporting exact linear readout of every query requires dimension at least $\text{rank}(F)$. For all size- s parity queries on d Boolean variables this becomes $\text{binom}(d, s)$ accessible coordinates, whereas a query-conditioned construction needs only the coordinates selected by the current query. Frozen controlled studies produce universal/compiled representation ratios of $91\times$ – $1820\times$ and dense-decoder sample-threshold reductions from $4\times$ to more than $32\times$. A no-answer-laundering construction exposes 5–7 latent components rather than the final label and retains large gains. A preregistered sparse universal decoder **falsifies** the stronger claim that compilation retains at least a $4\times$ threshold advantage in both hostile cells, leaving $2\times$ and $4\times$ residual gaps. A fresh deterministic replication reproduces those residuals; a separately frozen deterministic single-thread ExtraTrees successor fails to reach 0.95 through $n=1024$ where compiled state reaches it at $n=64$. Exact workload laws quantify option debt when specialized state is retained without raw recoverability. The result is not universal dominance of compilation: **state construction determines where structural-search computation is paid**, and the placement can be measured jointly through accessible rank, downstream burden, construction cost and future optionality.

1 Introduction

A system can possess all information required for a task and still make that information expensive for a bounded learner or reasoner to use. Computationally usable information formalizes one version of this observation. Partial evaluation and knowledge compilation move work upstream. Materialized views trade preprocessing and storage for later query cost. Current agent systems retrieve, compress, summarize or restructure context before generation. Recent LLM evidence also shows that state design itself can materially change dynamic reasoning while model parameters remain fixed.

Those results make several weak novelty claims untenable. P11 does **not** claim that representation matters, that computation can create usable information, that query-conditioned memory is new, or that compression can reduce downstream cost. The unresolved question is:

When task-relevant structure can be discovered either while constructing state or later by a decoder/search process, where is the computation paid, how much downstream burden can be removed, and what future optionality is lost by specialization?

We call this view *state as computation*. A compiler $\mathcal{C}(\mathbf{R}, \mathbf{q})$ receives raw state $\mathbf{R}$ and query $\mathbf{q}$, constructs task-facing state, and hands it to a bounded downstream access mechanism. Compilation is never free: compiler operations, state bytes, training cost, cache/recovery cost, downstream samples/search, verifier/tool calls, latency and reproducible energy belong to one resource receipt.

The paper contributes accessible-rank theory, controlled compilation gaps, hostile decoder substitution and future-optionality laws. The strongest conclusion is mechanistic: **representation construction and downstream access are two loci at which structural-search work can be paid.**

2 Donor boundary and novelty residual

Predictive V-information already establishes that computational constraints change what information is usable and that computation can transform unusable information into usable information. Classical partial evaluation specializes programs to known inputs. Knowledge compilation and database materialization move computation upstream for later reuse. Feature selection and sparse models search for relevant coordinates inside a larger representation. Query-conditioned memory and retrieval condition state on the current task. Long-horizon context-compression systems explicitly optimize memory/performance trade-offs. Wong et al. provide direct evidence that state representation and construction can change LLM reasoning.

P11 subtracts all of those primitives.

The residual is a **joint placement account**:

raw state $\rightarrow$ construction work $\rightarrow$ task-facing state $\rightarrow$ decoder/search work $\rightarrow$ verified outcome

with a future horizon:

task-facing state + raw/cache policy $\rightarrow$ future query service or option debt.

The paper asks how accessible rank, downstream sample/search burden, upstream construction, cache/recovery and future-query coverage move together when the same underlying information is exposed differently.

3 Accessible-rank theory

Let $\mathbf{X}$ be a domain with distribution μ , and let $\mathbf{F} = \{f_1, \dots, f_N\}$ be query functions in $L_2(\mu)$. A fixed query-agnostic representation is $\phi: \mathbf{X} \rightarrow \mathbb{R}^m$. It supports exact linear query answering when, for every $\mathbf{q}$, some $\mathbf{w}_{\mathbf{q}}$ satisfies $f_{\mathbf{q}}(\mathbf{x}) = \langle \mathbf{w}_{\mathbf{q}}, \phi(\mathbf{x}) \rangle$ almost surely.

3.1 Query-family accessible-rank bound

If $\text{span}(\mathbf{F})$ has dimension $\mathbf{r}$, every fixed representation supporting exact linear readout of all queries has $m \geq \mathbf{r}$.

Proof. Every $f_{\mathbf{q}}$ lies in the span of the m coordinate functions of ϕ . That span has dimension at most m and must contain the $\mathbf{r}$ -dimensional span of $\mathbf{F}$; hence $m \geq \mathbf{r}$. The theorem is elementary linear algebra; its role here is as a systems resource boundary, not as a claim of mathematical novelty.

3.2 Approximate orthonormal frontier

For orthonormal $f_1, \dots, f_N$ and any m -dimensional linearly accessible subspace $\mathbf{U}$, Bessel's inequality gives $(1/N) \sum_{\mathbf{q}} \|f_{\mathbf{q}} - P_{\mathbf{U}} f_{\mathbf{q}}\|_2^2 \geq 1 - m/N$. This is an access-class statement,

not a lower bound on unrestricted nonlinear decoding.

3.3 Parity corollary

For $\mathbf{X}=\{-1,+1\}^d$ under the uniform measure and all size- s subsets S , define $f_S(\mathbf{x})=\prod_{i \in S} x_i$. Distinct parity characters are orthogonal. A fixed exact linear-accessible representation supporting all size- s queries therefore requires at least $\text{binom}(d,s)$ coordinates. A query-conditioned construction need expose only the selected query structure. This establishes an exact accessible-representation gap, not a total-time lower bound.

4 Dense controlled studies

The confirmatory experiment compares raw linear input, a fixed universal parity bank, and query-conditioned compiled state over a frozen train-size grid.

d	s	universal dims	compiled dims	universal/compiled	compiled n at 0.90	universal n at 0.90
14	2	91	1	91×	32	128
16	4	1820	1	1820×	32	NOT_REACHED by 1024
18	3	816	1	816×	32	1024
20	3	1140	1	1140×	32	1024

At $n=1024$, compiled accuracy is 1.0 in every cell. Raw linear remains near chance. The result is consistent with a structural-search interpretation: decisive coordinates exist in the universal bank, but a bounded dense decoder must identify them among many irrelevant candidates.

4.1 No-answer-laundering control

P11B exposes only the r active parity components selected by a query, with r in $\{5,7\}$. The downstream logistic learner must still infer an odd-cardinality majority target; no compiled component equals or negates the final label.

d	s	r	universal dims	universal n at 0.95	compiled n at 0.95
15	3	5	455	2048	64
17	3	5	680	2048	64
17	4	5	2380	NOT_REACHED by 2048	64
19	3	7	969	NOT_REACHED by 2048	64

The high-dimensional cells therefore exhibit at least $32\times$ threshold separation under the registered dense decoder without answer laundering.

5 Hostile decoder substitution

The central alternative explanation is that the universal representation is penalized only because the downstream decoder has the wrong inductive bias. If so, stronger decoder-side search should buy back the compilation advantage. P11 treats that prediction as a mechanism test.

5.1 P11D sparse decoder — permanent negative

P11D preregistered a strong hostile gate: an L1 sparse universal decoder should still leave at least a $4\times$ sample-threshold advantage for compiled state in both high-dimensional cells. It did not.

cell (d,s,r)	sparse universal n at 0.95	compiled n at 0.95	ratio	compiled - sparse at n=64
(17,4,5)	128	64	2×	+0.2903
(19,3,7)	256	64	4×	+0.3840

The terminal `P11D_SPARSE_DECODER_GAP_NOT_MET` remains permanently negative. P11D also exposed an unseeded `liblinear` replay defect; that defect is retained.

5.2 P11E deterministic sparse replication

P11E uses a fresh data seed and explicit estimator seeds. It reproduces thresholds 128/64 and 256/64 with low-sample compiled-minus-sparse advantages +0.2912 and +0.3307. Two fresh executions produce canonical SHA-256 `1097d94bef1132d4dfa5d01176a9fcfcfebc46de8113e7cb2e57da1e579a4536`.

5.3 P11C/P11F/P11G nonlinear sequence

P11C emitted no authoritative terminal and remains `CANNOT_CHECK`. P11F produced positive numerical separation but is non-authoritative because hostile review found `n_jobs=-1` violated the frozen otherwise-default tree contract. P11G was frozen separately with `n_jobs=1`, explicit random states, and two fresh subprocess replays in the terminal decision path.

cell	deterministic tree universal n at 0.95	compiled n at 0.95	tree accuracy at n=1024	compiled -
(17,4,5)	NOT_REACHED	64	0.8248	
(19,3,7)	NOT_REACHED	64	0.7828	

P11G's terminal is `P11G_DETERMINISTIC_TREE_DECODER_GAP_SUPPORTED`; both fresh scientific payloads have SHA-256 `a2b0c33ce3c39e54ca1aa400a2b7d52d019fc4503f6cd5eb726c7b8bbe79a7cc`.

The sequence supports the interpretation that **compilation and decoder inductive bias are alternative locations for structural search**. Stronger downstream structure discovery should shrink the upstream advantage; the sparse negative is therefore part of the mechanism evidence, not an inconvenient result to erase.

6 Future optionality and accessibility-work accounting

Current accessibility and future optionality are different objectives. Let a universal state contain N independent query coordinates and a compiled state retain r .

If raw source is lost, one-step coverage for a uniformly random future query is exactly r/N . If K independently selected size- r compilations are cached, expected coverage is $1-(1-r/N)^K$. Under uniform demand, the expected number of distinct requested components after K queries is $N[1-(1-r/N)^K]$.

These laws produce four policy regimes: compile only; compile + cache; retain raw + compile; and universal materialization. In the frozen workload study, the first grid point where universal bulk compilation becomes cheaper than expected cache compilation occurs at $0.5N$, $1.0N$, and $2.0N$ future-query horizons for batch-efficiency factors 0.25, 0.50, and 0.75. Concentrated Zipf-like workloads delay crossover because fewer distinct coordinates are demanded.

Any comparison that treats representation construction as free is invalid. P11 therefore keeps separate receipt coordinates for compiler/preprocessing operations and latency, state bytes/tokens and memory traffic, downstream model identity/capacity, training examples or generated/recurrent steps, search nodes/verifier/tool calls, cache and reuse, raw recovery/reconstruction/recompilation, end-to-end latency and reproducible energy where available.

Learned-compiler training cost must be reported separately and amortized only under a prospectively stated reuse horizon. Unless an application supplies exchange rates, comparisons should be Pareto surfaces rather than post-hoc scalar costs.

7 Methods, reproducibility and related work

The controlled experiments use prospectively frozen seeds, cell grids, train-size grids, thresholds and hostile checks. Thresholds are observed first grid points and are never extrapolated beyond the registered maximum; `NOT_REACHED` is retained literally. Exact theorems and workload equations require no statistical inference.

The evidence history is append-only in scientific meaning: P11/P11B establish dense and no-answer-laundering gaps; P11D fails its stronger sparse gate; P11E independently reproduces the surviving residual; P11C remains `CANNOT_CHECK`; P11F is diagnostic after a protocol-conformance defect; P11G is separately frozen with replay as a hard terminal gate.

Predictive V-information is the closest information-theoretic parent. Partial evaluation and knowledge compilation are the closest upstream-computation parents. Materialized views and multi-query optimization are reuse/crossover analogues. Modern query-conditioned memory, retrieval and context compression demonstrate practical task-conditioned state construction. Wong et al. provide current evidence that state design changes dynamic LLM reasoning. Sparse estimators and nonlinear ensembles are natural downstream substitutes.

P11’s residual is not any primitive in isolation. It is the experimentally attacked relation between **where structure is exposed, how much discovery work remains downstream, what that exposure costs, and what future options it destroys or preserves.**

8 Limitations, discussion and conclusion

The exact theorem is restricted to a declared linear-access class and is not an unrestricted representation or time lower bound. Parity families are deliberately controlled. P11D shows that stronger decoder bias materially reduces compilation advantage. P11G tests one finite nonlinear ensemble. The current compiler is oracle/query-structured rather than learned. Controlled operation counts do not replace end-to-end latency, memory traffic, training cost or energy. Future-query laws assume the declared workload model. Broad real-system superiority still requires matched same-information experiments in real LLM/procedural and formal/search or long-horizon settings.

The experiments change the interpretation of task-conditioned construction. The strongest story is not that a compiler creates information or universally beats a universal representation. The decisive variable is **who performs structural search**. A query-conditioned compiler performs some search before the downstream learner sees state. A sparse decoder can recover part of that work; a deterministic finite tree ensemble is another search mechanism but remains inefficient under the registered high-dimensional low-sample envelope.

Aggressive specialization can make one task cheap while destroying immediate support for future tasks. Retaining raw state, caching compilations or materializing wider state are different allocations of computation and memory across time.

P11 establishes a controlled theory/systems result: fixed linear-accessible state scales with query-family rank; task-conditioned construction can expose smaller task-facing state; dense-decoder experiments show large gains; a hostile sparse decoder buys part of the gain back but leaves a reproducible $2 \times / 4 \times$ residual; and a replay-gated deterministic nonlinear tree ensemble does not close the high-dimensional gap under its registered budget.

State construction, decoder search, and future recoverability are coupled resource choices. Moving structure into state can reduce downstream discovery work, but the benefit shrinks as the downstream access mechanism becomes better at discovering that structure, and specialization creates option debt unless recoverability is retained.

The controlled claim does not authorize a universal nonlinear lower bound, transformer scaling law, free preprocessing or real-agent superiority.

9 References

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